

Hi, I'm trying to create a web-based application that will operate as an independent identity taking Meal and medicine administration input from support workers systematically and save them dynamically.

Also I want to make Two engines that generates these tables dynamically taking various Parameters from the user but the output should look like its audit friendly

At first lets set the conventional app where the data entry will take place and systematically saved in databases

And secondly the additional engine we are building is separate from the logic; but it does data outputs overriding our will populate the database faster incase of disaster management (audit defense)

My application has two generators. The first intake engine for (Mealtime) has 9 headers ; I will at first set the data types with their respective range and constraint

Meal Chart Data Constraints

Header Field	Data Type (Domain)	Range / Constraints (Values)
Entry ID	UUID / String	Range: Unique identifier for each individual log (e.g., Log_123).
Participant ID	String	Range: Must match a valid entry in the participant_profile table.
Date & Time	Timestamp	Range: ISO 8601. Auto-generated at "Submit." Must be

		Current_Time.
Type of Meal	Enum	Range: [Breakfast, Morning Tea, Lunch, Afternoon Tea, Dinner, Snack, Fluid Only].
Food Texture Level	Integer	Range: 3 to 7 (IDDSI Food Standards). Pulls default from Participant Plan.
Fluid Thickness Level	Integer	Range: 0 to 4 (IDDSI Drink Standards). Pulls default from Participant Plan.
Fluid Volume	Integer	Range: 0 to 2000 (ml). Numerical input for dehydration tracking.
Amount Eaten	Enum	Range: [0%, 25%, 50%, 75%, 100%]. Tracks nutritional intake.
Position Used	String	Range: e.g., "Upright 90 degrees," "In wheelchair with head support."
Swallowing Obs	Array / Text	Range: Checkboxes: [None, Coughing, Choking, Gurgly Voice, Drooling].
Worker ID	String	Range: Must match an authenticated staff_id currently logged in.

Generator Logic (Mealtime) for the DB input :

Incase we are manually putting all the values from scratch ; This engine will prompt the user for certain parameter to fully automatically populate all the fields

First input : from where do you want to start your entry_ID or LogID from?
Once the Input is given it will automatically Populate where needed

2ndly for bulk chart for a certain user

Header Field	Data Type (Domain)	Generator Logic
Entry ID	UUID / String	Entry ID will increment after one entry ++
Participant ID	String	Remains same for individual participant
Date & Time	Timestamp	(for bulk generation; we will use random function but with a range of time ; once range of time is put for N number of entries ; it will randomly select a time and second to make it look real)
Type of Meal	Enum	According to the range: [Breakfast, Morning Tea, Lunch, Afternoon Tea, Dinner, Snack, Fluid Only]. When we put entries

Food Texture Level	Integer	Here we will use a dynamic function for bulk update [GPT we need to brainstorm]
Fluid Thickness Level	Integer	Linked with the row ; this is also a dynamic function will work according to the Type of meal , texture
Fluid Volume	Integer	We will put BIAS/ or range and random function to make it look authentic ; but since Its drink we are making usually the standard amount so No need to put decimal or randomness
Amount Eaten	Enum	Amount eaten is also random() but BIAS cause we usually finish the food
Position Used	String	According to the range ; It mostly depends on what we are eating but We usually eat upright and clients has choking risk so we always adhere to standard
Swallowing Obs	Array / Text	Range checkboxes; usually dont happene but we will put manually not in bulk generator
Worker ID	String	For bulk we will set up dates backdated, the worker data will be pulled from the SaaS using selenium extractor ; The worker ID not only gives us Unique Worker Identifier

Logic & Validation Rules for your Developers:

- **The "None" Logic:** In the Swallowing Obs field, if the worker selects **"None"**, the app should automatically deselect any other risk factors (like "Coughing") to prevent logical data errors.
- **The IDDSI Validation:** If the Food Texture Level entered by the worker is **different** from the level prescribed in the Participant Plan, the app should trigger a mandatory **"Reason for Deviation"** text box.
- **Volume Limits:** Set a "Soft Warning" in the UI if Fluid Volume exceeds **1000ml** in a single entry to catch potential typing errors (e.g., accidentally typing 2500 instead of 250).

Example dataset with headers :

```
{  
  "entry_id": "LOG-123",  
  "participant_id": "NDIS-789-XY",  
  "timestamp": "2026-02-19T13:45:00Z",  
  "data": {  
    "meal_type": "Lunch",  
    "actual_food_texture": 4,  
    "actual_fluid_thickness": 2,  
    "volume_ml": 200,  
    "percent_eaten": 75,  
    "positioning": "Upright 90 degrees",  
    "swallowing_observations": ["Coughing"],  
    "staff_id": "STAFF-55"  
  }  
}
```

Each field has unique data types for each case.

We have to match them accordingly

so initially there are two data types.

I haven't decided yet.

My ultimate goal is to generate Mealtime Charts and Medication Charts.

When I provide the parameters, I expect a very usable UI.

I'm guessing we can use a web browser to view it or create the UI using modern libraries.

Medicine Chart (MAR) Data Constraints

Header Field	Data Type (Domain)	Range / Constraints (Values)
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Participant ID	UUID / String	Range: Must match a valid entry in the participant_profile table.
Scheduled Time	Time	Range: 00:00:00 to 23:59:59. Usually matches the "Prescribed Time" (e.g., 08:00 AM).
Actual Admin Time	Timestamp	Range: ISO 8601. Must be Current_Time. Captures the exact moment of administration.
Medication Name	String	Range: 1–255 chars. Requirement: Use the Active Ingredient name (e.g., Paracetamol) as per NDIS guidelines.
Dose	String	Range: Must include quantity and unit (e.g., "50mg", "2 tablets", "5ml").
Route	Enum	Range: [Oral, Topical, Inhaled, Injection, PR (Rectal), Ear Drops, Eye Drops].
Status	Enum	Range: [Administered, Refused, Omitted, Self-Administered, Nauseous/Vomited, Leave/Away].
Omission Reason	String	Range: Null if status is "Administered"; Mandatory if status is "Refused" or "Omitted".
PRN Indication	String	Range: Required ONLY if the medication is "As Required" (e.g., "Reported headache").
Worker Signature ID	String	Range: Must match an authenticated staff_id.

Example dataset with Headers:

```
{
  "record_type": "medication_administration",
  "participant_id": "NDIS-12345-AB",
  "medication_id": "MED-SER-50",
  "prescription": {
    "drug_name": "Sertraline (Zoloft)",
    "dosage": "50mg",
    "route": "Oral",
    "prescribing_doctor": "Dr. Sarah Smith",
    "start_date": "2026-01-15"
    "expiry_date": "2027-01-15"
  },
  "administration_event": {
    "scheduled_time": "08:00:00",
    "actual_admin_datetime": "2026-02-19T08:12:45Z",
    "status": "administered",
    "omission_reason": null,
    "administrator_id": "STAFF-987",
    "administrator_signature_thumbprint": "sha256-hash-value"
  },
  "observations": {
    "side_effects_noted": false,
    "notes": "Participant took medication with water, no issues."
  },
}
```



```
"audit_metadata": {  
  "created_at": "2026-02-19T08:13:00Z",  
  "is_immutable": true,  
  "version": 1.0  
}  
}
```